QuantForge

多模式自適應量化策略 完整研究報告

4 定價模型 (BSM / Bates / Heston / SVI) × 11 標的 × 10 年

2026-04-30

GOOGL AAPL MSFT NVDA SPY QQQ

AMZN TSLA JPM JNJ XOM

目錄

目錄

1. 策略架構 ..... 4 模式決策樹 + 並行引擎說明
2. 定價模型介紹 ..... BSM / Bates / Heston / SVI 原理
3. 定價模型對比表 ..... 每支股票 x 4 模型的 Wheel 收益比較
4. 定價模型影響圖 ..... Premium 差異分組柱狀圖
5. 最佳模型策略總結 ..... 各標的最佳模型及績效
6. 全標的權益曲線 ..... Baseline vs MultiMode
7. 模式分佈圖 ..... 堆疊柱狀圖
8. 逐標的權益對比 ..... 11 張個股圖 (含模式色帶)
9. Wheel 詳情頁 ..... CSP/CC 週期、指派、權利金
10. 關鍵發現與建議 ..... 定價模型結論 + 策略結論

策略架構

多模式決策樹

每根 K 棒根據市場狀態(Regime)和年化波動率選擇模式：

```
BEAR / CRISIS        -> CASH       (不交易)
BULL + vol > 45%      -> HIGHVOL_BREAKOUT (4x ATR 追蹤停損)
BULL + vol <= 45%     -> TREND_FOLLOW    (3x ATR 追蹤停損)
CONSOLIDATION / NEUTRAL -> WHEEL         (CSP/CC 選擇權收入)
```

並行引擎

三個引擎同時運行：
1. 趨勢/突破引擎 — 多個同時持有的趨勢/突破部位
2. Wheel 引擎　— 模擬 Cash-Secured Put / Covered Call 週期
3. 現金引擎　— 熊市/危機期間不開新倉

Regime-Adaptive Sizing

```
RegimeDetector:  VIX > 30 = CRISIS, ADX + MA60 判斷趨勢
部位大小依 regime 調整：
  Bull:    50% base, 10% 風險上限
  Neutral: 30% base,  5% 風險上限
  Consol:  20% base,  3% 風險上限

最大總曝險: 80%
持有上限:  250 交易日
```

定價模型介紹

BSM (Black-Scholes-Merton)

經典歐式選擇權定價。假設波動率恆定、無跳躍。
優點：計算快速、美元定價最準 (vega 加權 IV RMSE 0.0627)
缺點：無法捕捉波動率微笑和跳躍風險

Bates (跳躍擴散模型)

Merton 跳躍擴散封閉解：BSM 價格 x Poisson 跳躍調整項
參數：lambda=0.1 (跳躍頻率), muJ=-0.05 (跳躍均值),
sigmaJ=0.10 (跳躍波動率)
優點：液態選擇權上最佳 (IV RMSE 0.0549)
特性：premium >= BSM (跳躍項增加尾部風險)

Heston (隨機波動率近似)

二次偏態近似：adjusted_vol = v0 + rho*xi/(2*v0) * moneyness
參數：kappa=2.0, theta=vol^2, xi=0.3, rho=-0.7
優點：速度快 (一次 BSM 呼叫)
缺點：近似精度一般，介於 BSM 和 SVI 之間

SVI (Stochastic Volatility Inspired)

Gatheral SVI: w(k) = a + b*(rho*(k-m) + sqrt((k-m)^2 + sigma^2))
k = ln(K/S), w = implied_var * T
參數：a=0.04, b=0.1, rho=-0.4, m=0, sigma=0.1
優點：波動率預測最好 (OOS IV RMSE 0.163)
缺點：美元定價最差

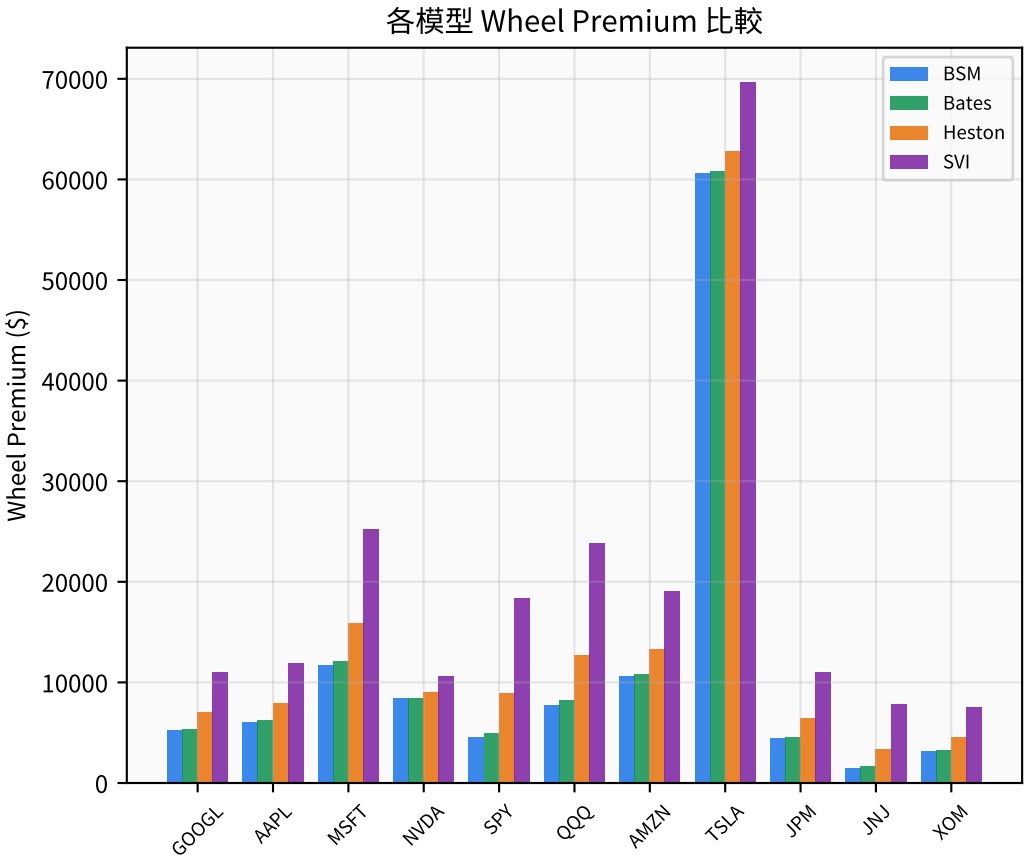
研究結論摘要

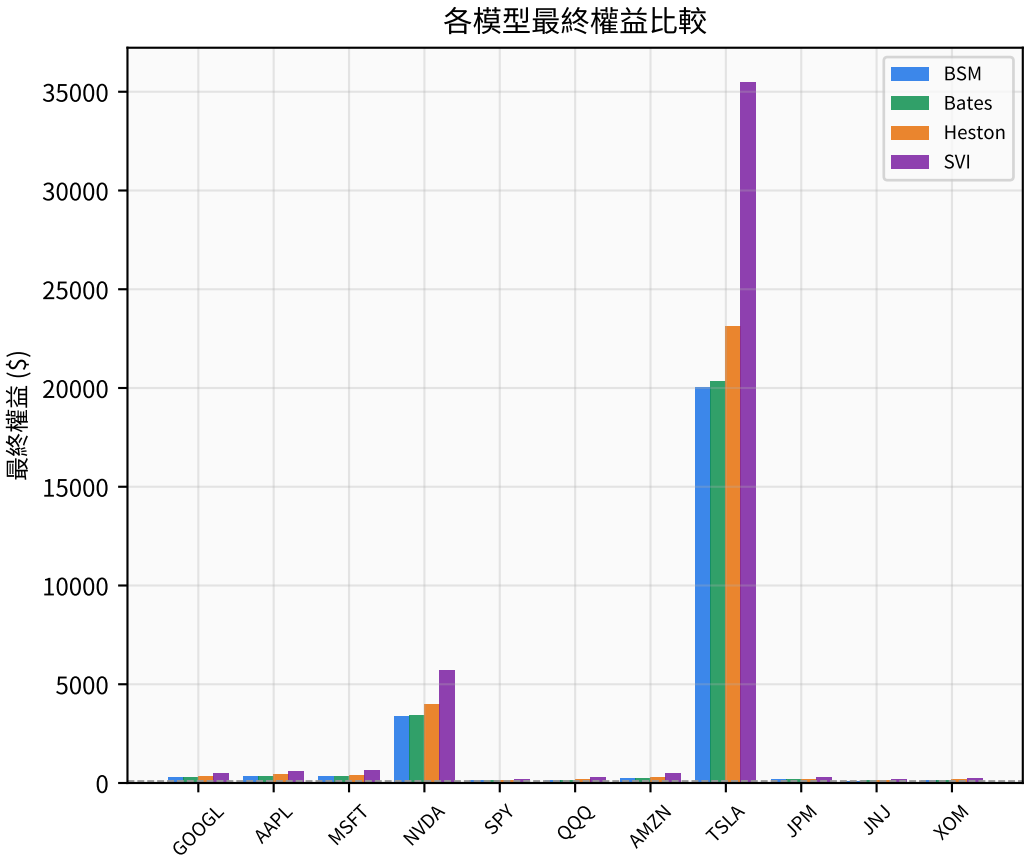
模型	IV RMSE	美元精度	速度	最佳場景
BSM	0.0627	最佳	極快	一般用途
Bates	0.0549	次佳	中等	液態期權
Heston	中等	中等	快	偏態定價
SVI	0.163	最差	快	波動率預測

定價模型對比 — Wheel Premium 及最終權益

標的	BSM Premium	Bates Premium	Heston Premium	SVI Premium	BSM Eq	Bates Eq	Heston Eq	SVI Eq
GOOGL	$5197	$5357	$7039	$10995	$268	$273	$324	$488
AAPL	$6004	$6166	$7861	$11835	$344	$350	$407	$586
MSFT	$11683	$12076	$15874	$25204	$320	$326	$392	$615
NVDA	$8375	$8442	$8977	$10576	$3380	$3429	$3957	$5721
SPY	$4512	$4970	$8868	$18314	$114	$116	$131	$172
QQQ	$7706	$8201	$12689	$23795	$144	$146	$172	$256
AMZN	$10595	$10824	$13237	$19004	$245	$250	$301	$475
TSLA	$60546	$60800	$62808	$69609	$20028	$20351	$23115	$35456
JPM	$4376	$4571	$6442	$10944	$169	$172	$202	$298
JNJ	$1479	$1685	$3370	$7834	$101	$102	$117	$169
XOM	$3109	$3234	$4501	$7517	$131	$134	$161	$250

定價模型影響 — Premium 及權益比較

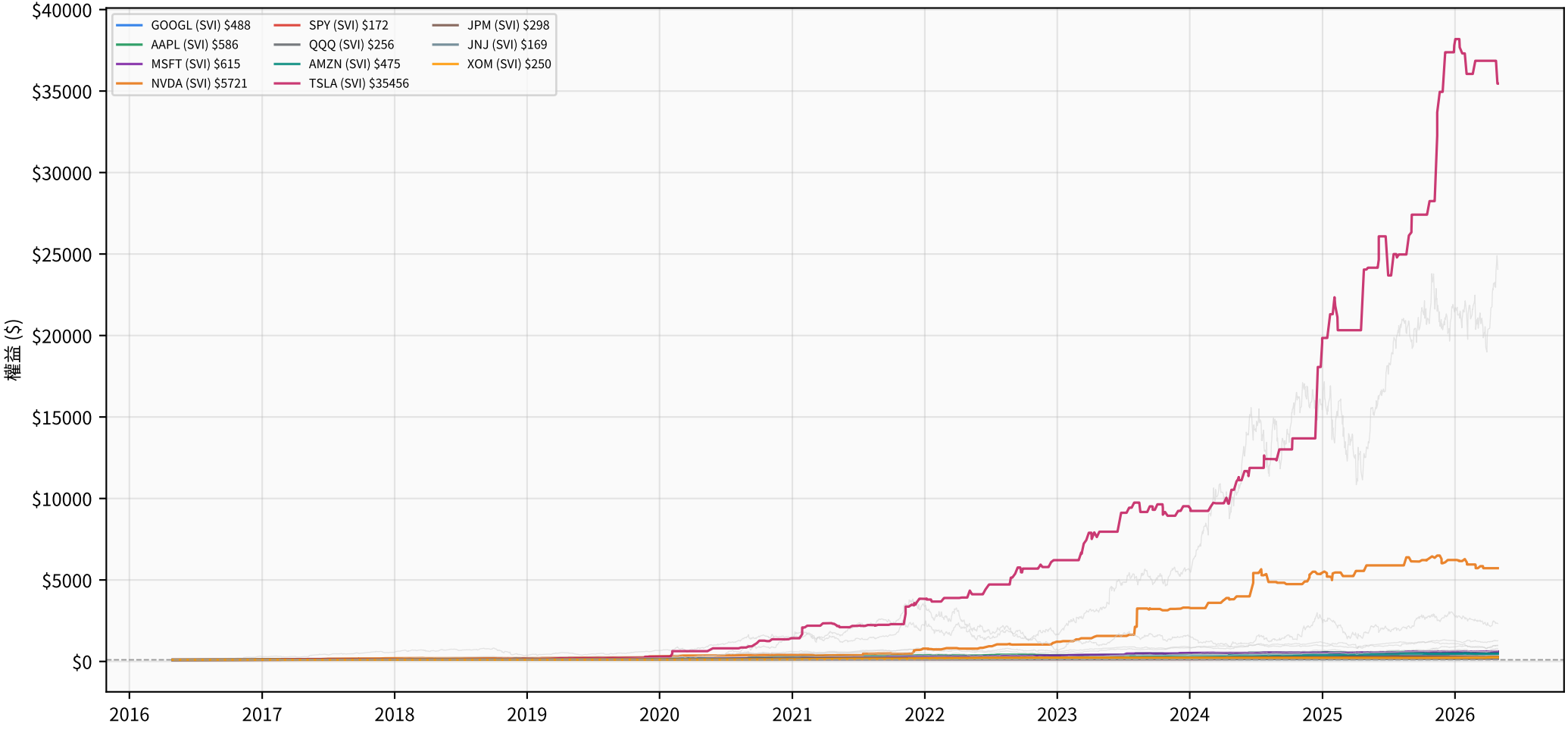




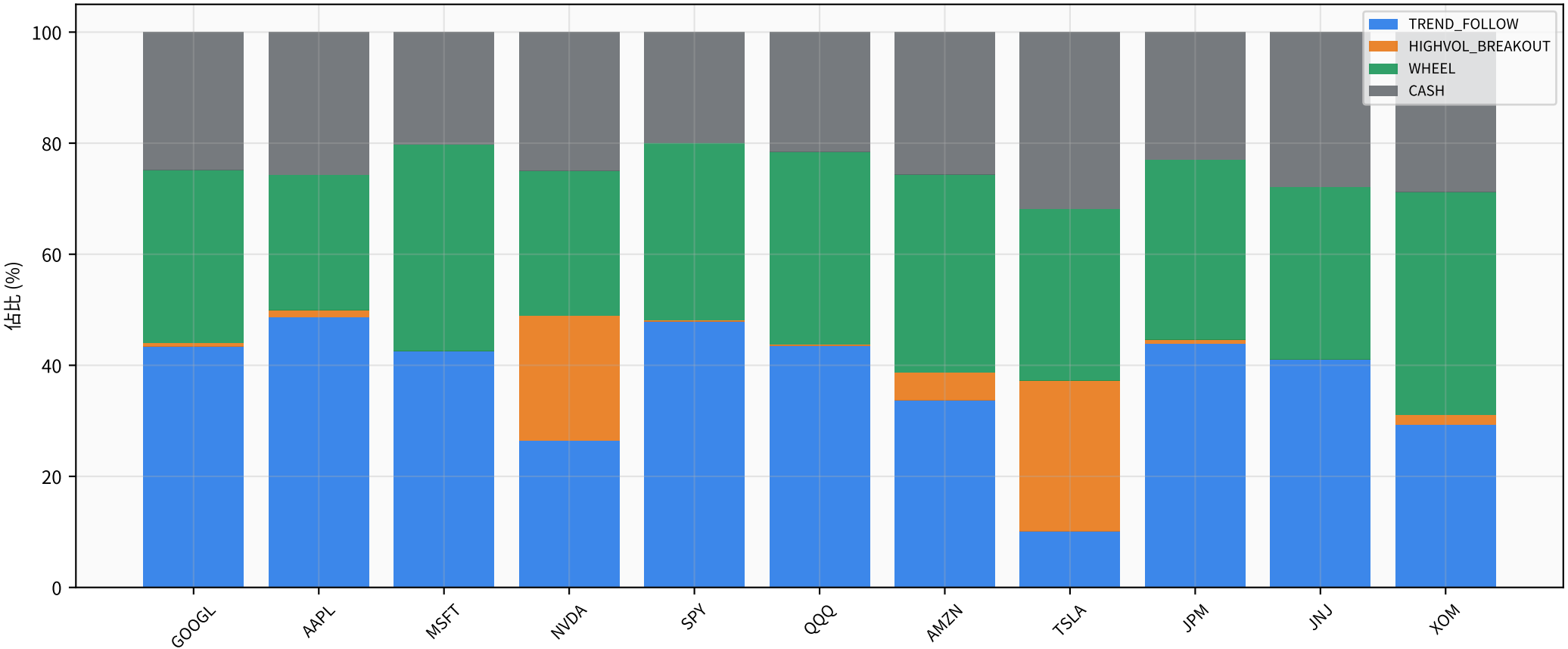
最佳模型策略總結 — 各標的績效

標的	Baseline $	最佳模型	最佳 Eq $	Sharpe	Wheel Premium	CSP	CC	指派	主要模式
GOOGL	$994	SVI	$488	2.98	$10995	68	2	3	TREND_FOLLOW
AAPL	$1274	SVI	$586	3.03	$11835	61	1	1	TREND_FOLLOW
MSFT	$965	SVI	$615	3.47	$25204	77	2	3	TREND_FOLLOW
NVDA	$24055	SVI	$5721	3.18	$10576	63	8	10	TREND_FOLLOW
SPY	$405	SVI	$172	1.90	$18314	57	0	1	TREND_FOLLOW
QQQ	$669	SVI	$256	2.56	$23795	73	0	2	TREND_FOLLOW
AMZN	$796	SVI	$475	2.77	$19004	75	4	5	WHEEL
TSLA	$2321	SVI	$35456	3.75	$69609	74	18	19	CASH
JPM	$637	SVI	$298	2.31	$10944	69	0	1	TREND_FOLLOW
JNJ	$268	SVI	$169	1.71	$7834	68	1	1	TREND_FOLLOW
XOM	$270	SVI	$250	2.19	$7517	76	2	3	WHEEL

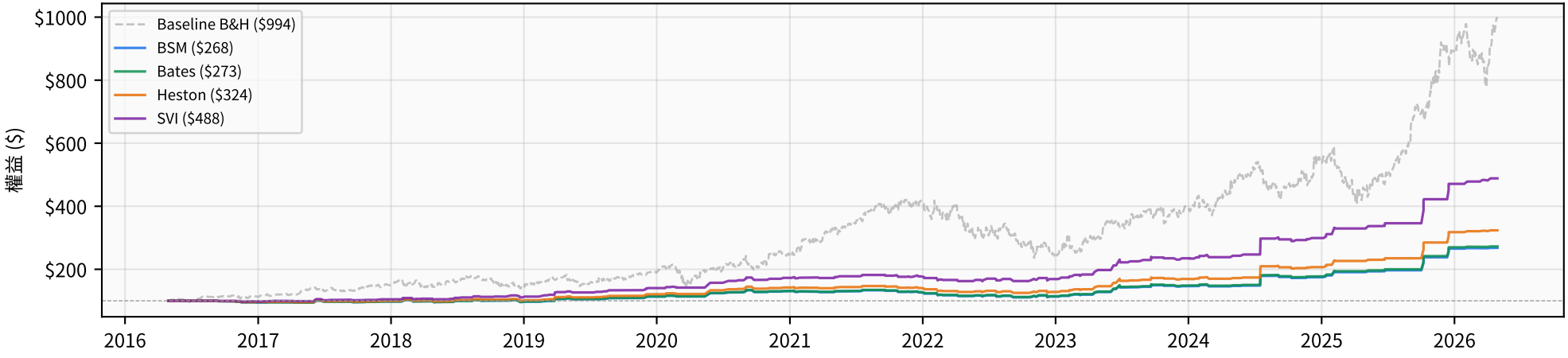
全標的權益曲線 — Baseline vs 最佳 MultiMode

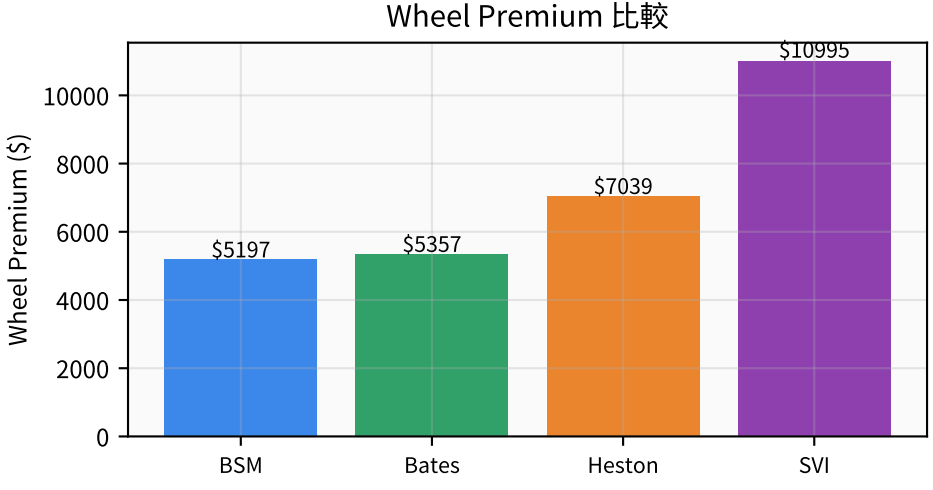


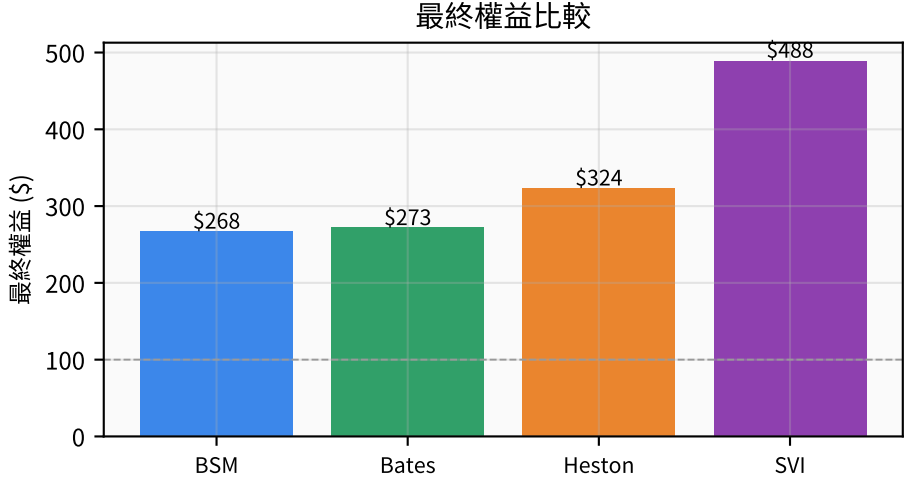
模式分佈 — 各標的模式時間佔比 (最佳定價模型)



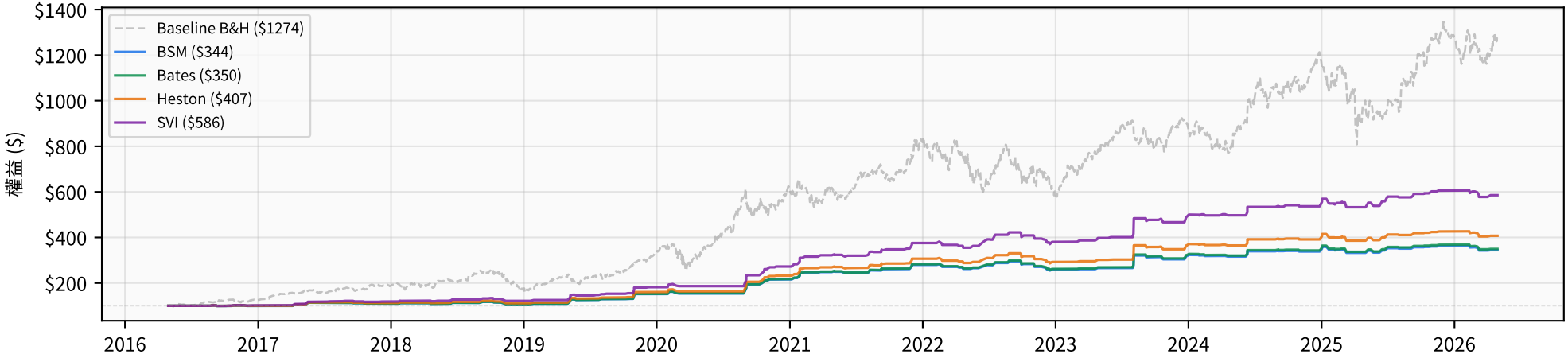
GOOGL — 4 定價模型權益對比 ($100 起始)

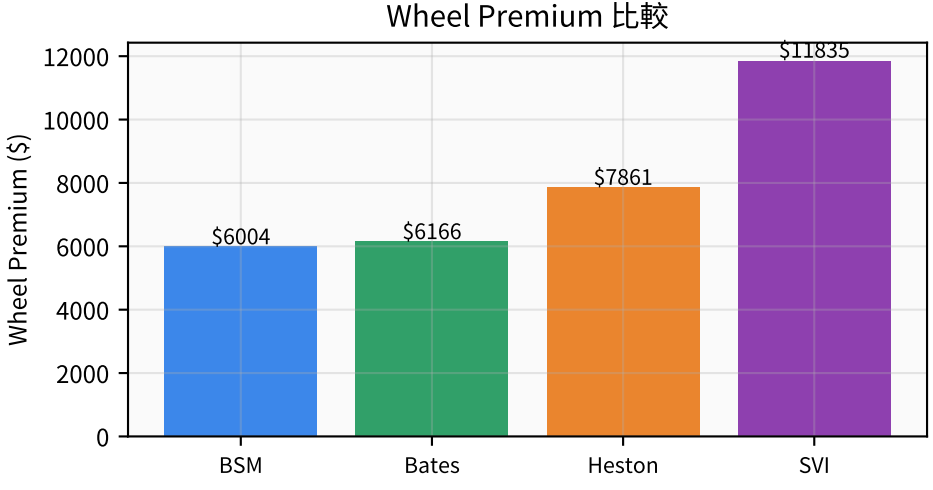


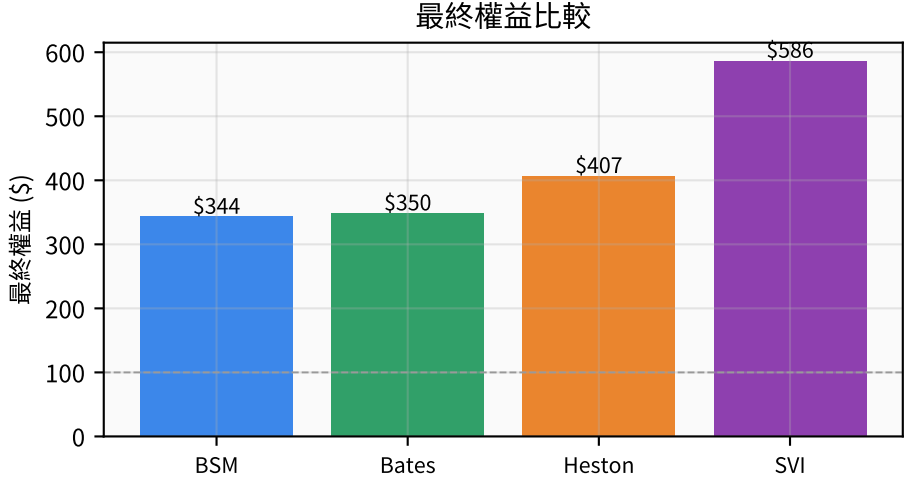




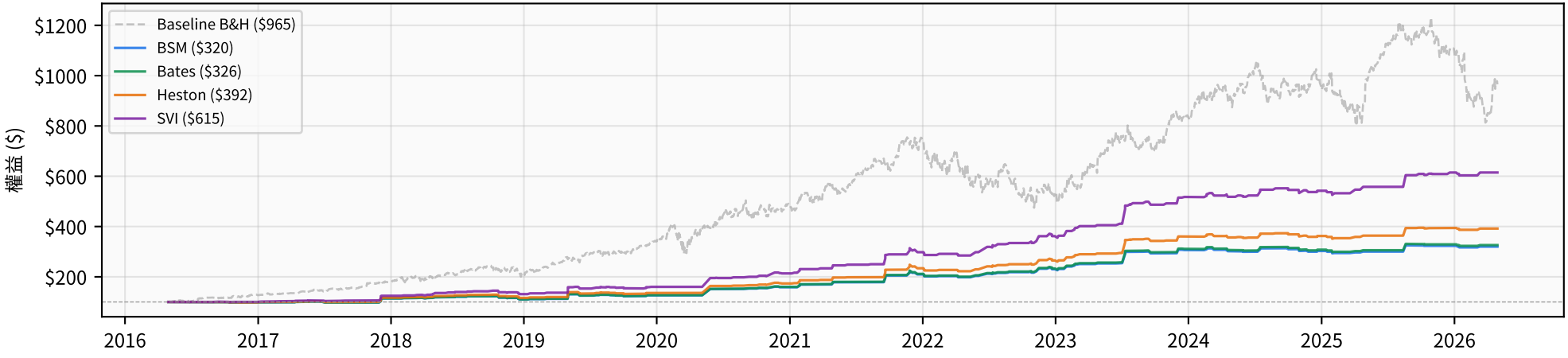
AAPL — 4 定價模型權益對比 ($100 起始)

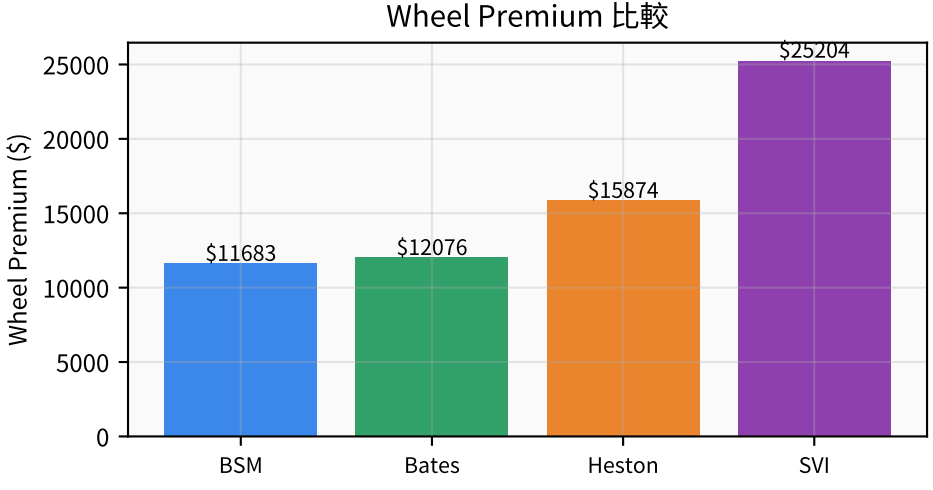


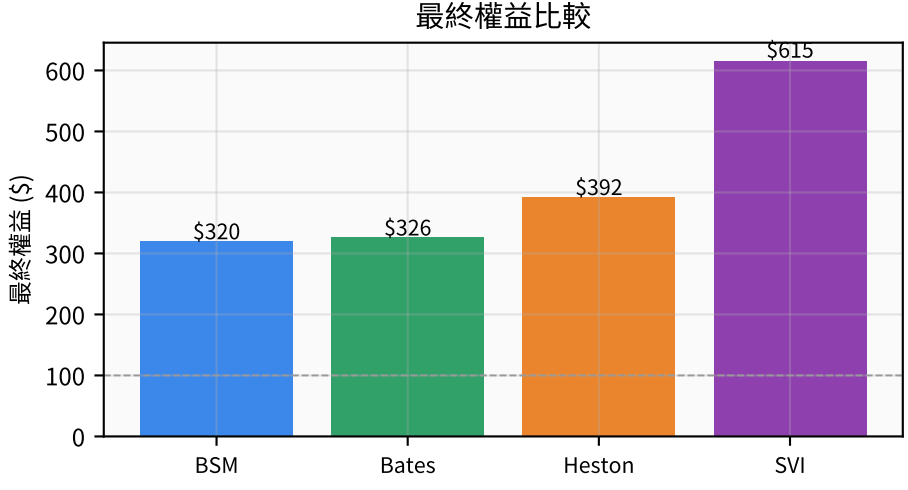




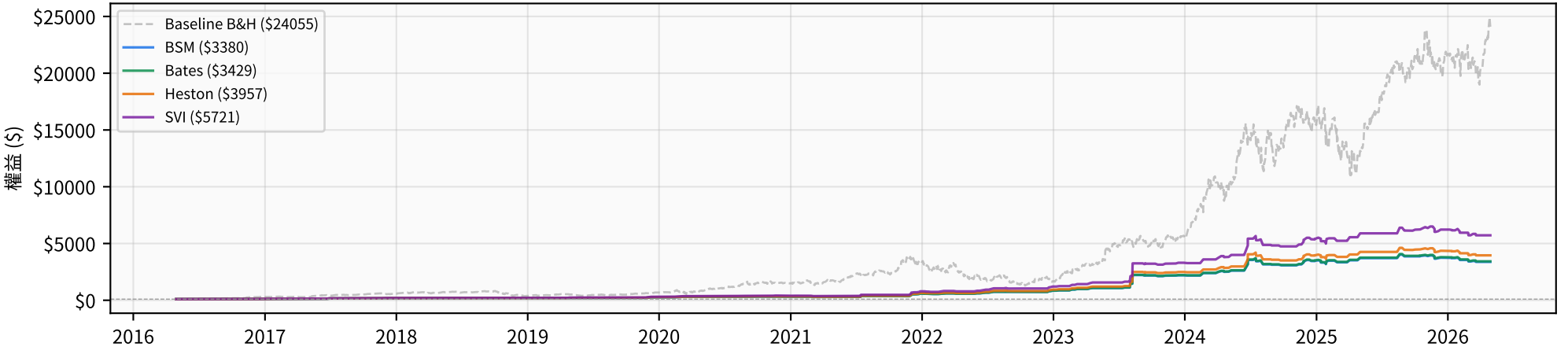
MSFT — 4 定價模型權益對比 ($100 起始)

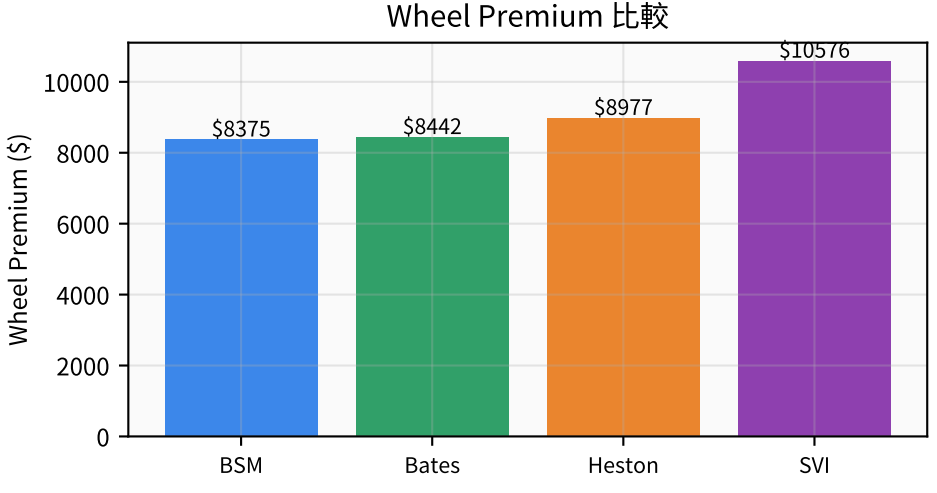


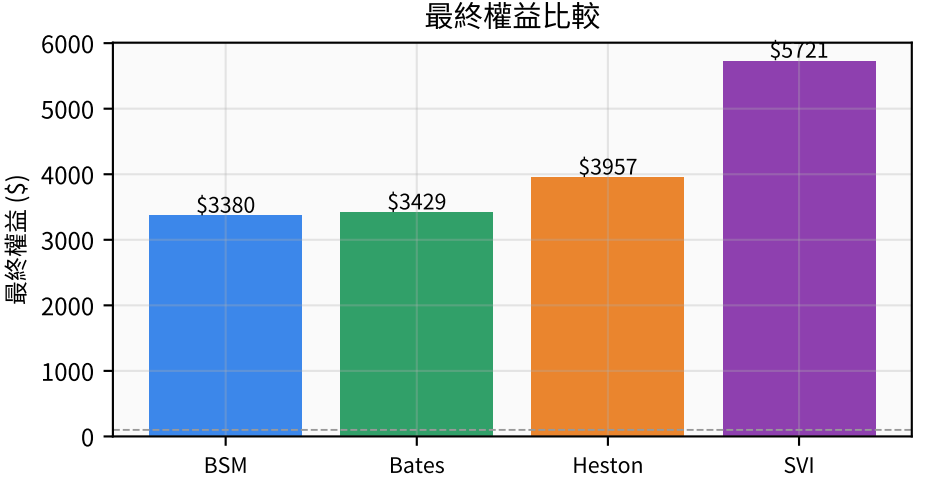




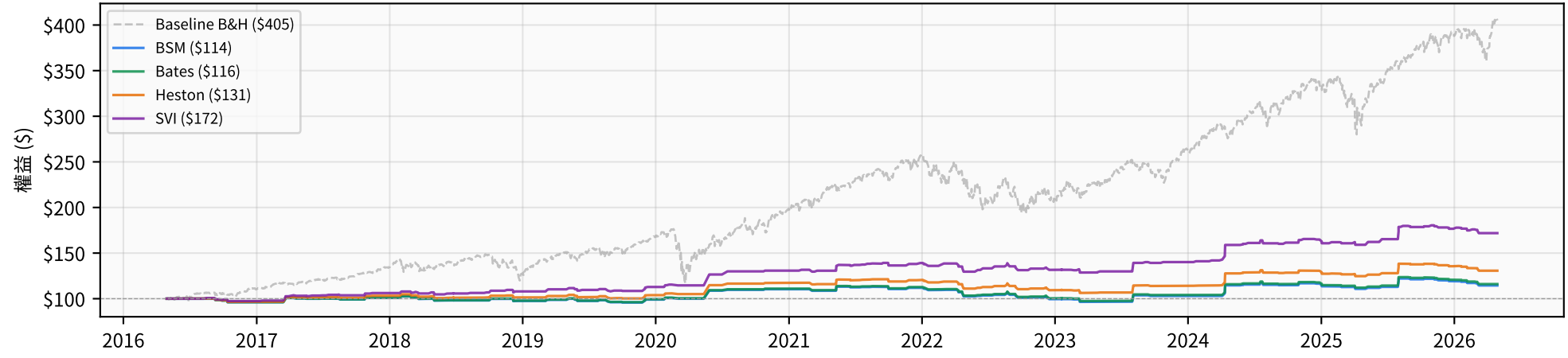
NVDA — 4 定價模型權益對比 ($100 起始)

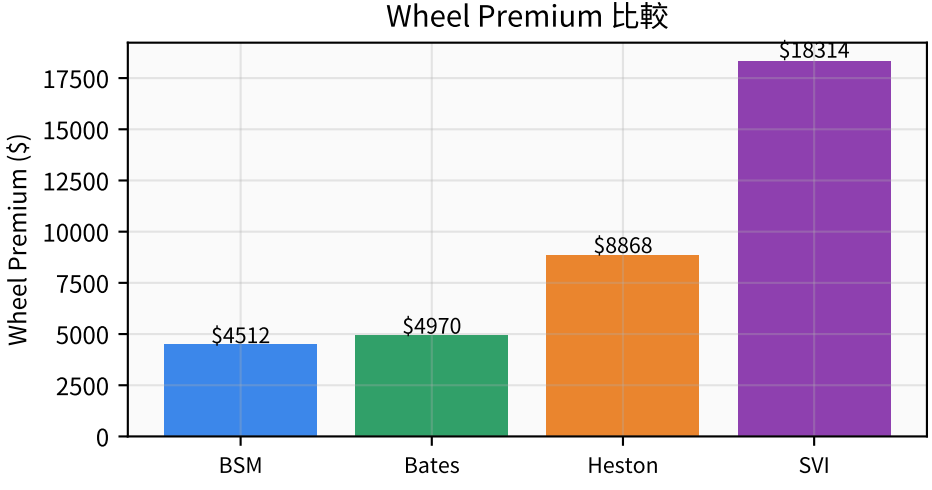


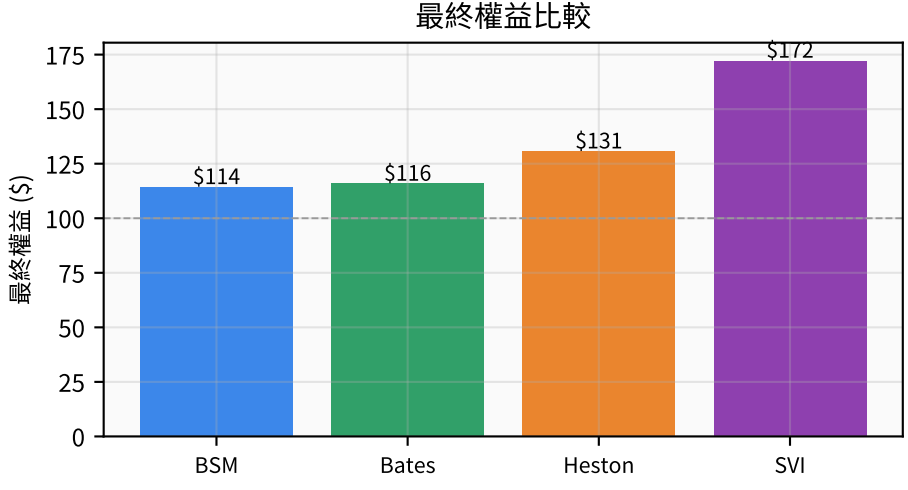




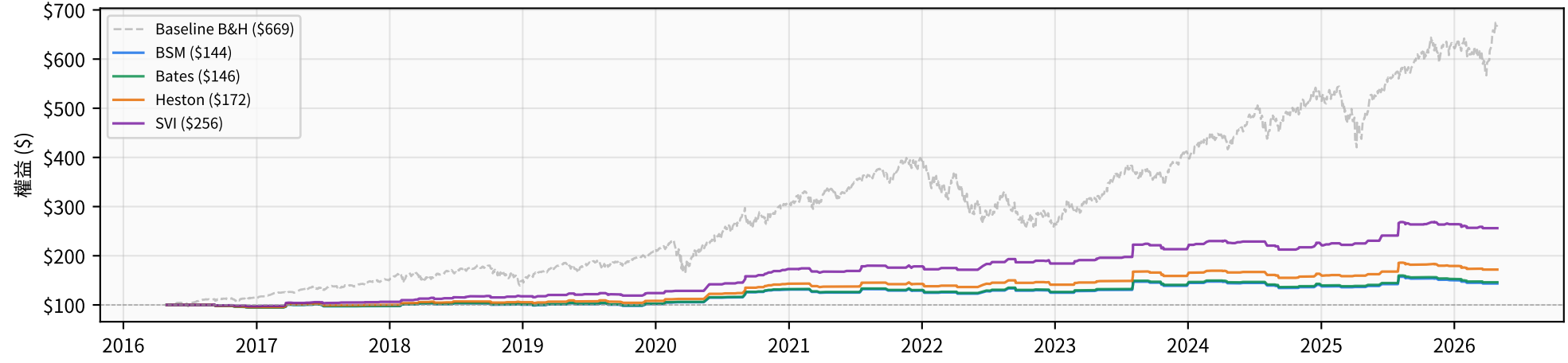
SPY — 4 定價模型權益對比 ($100 起始)

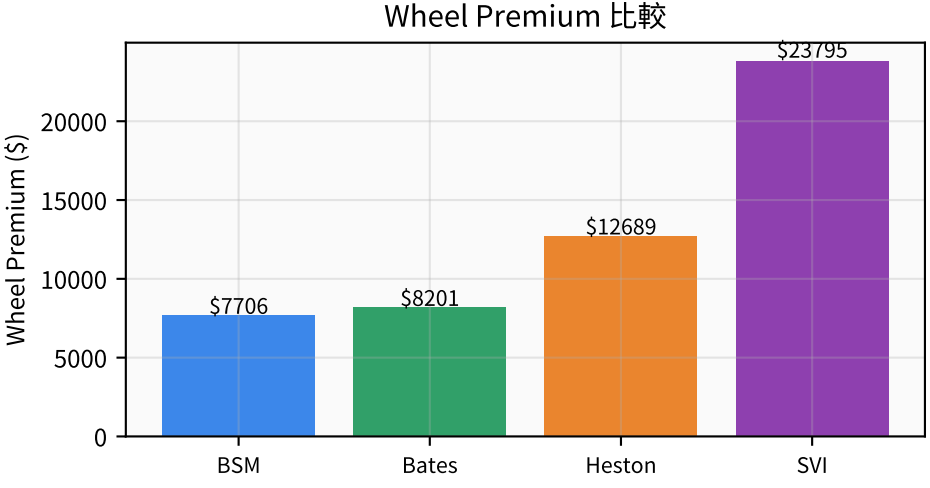


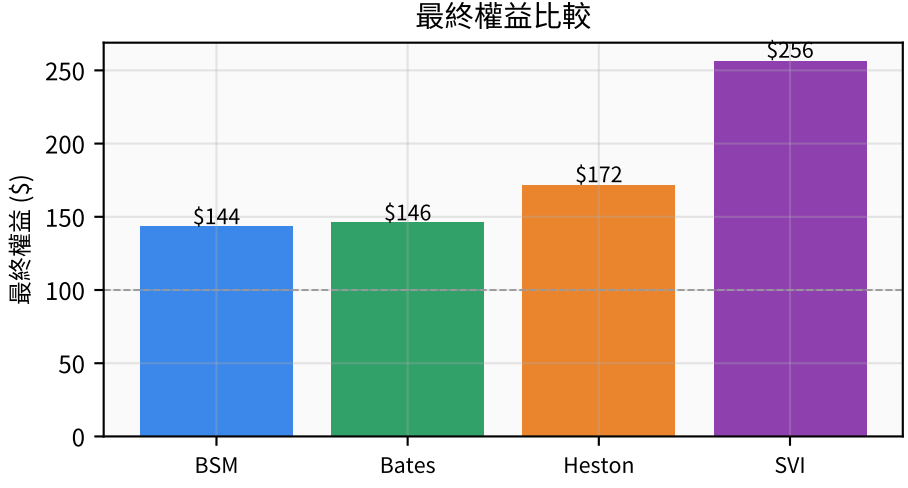




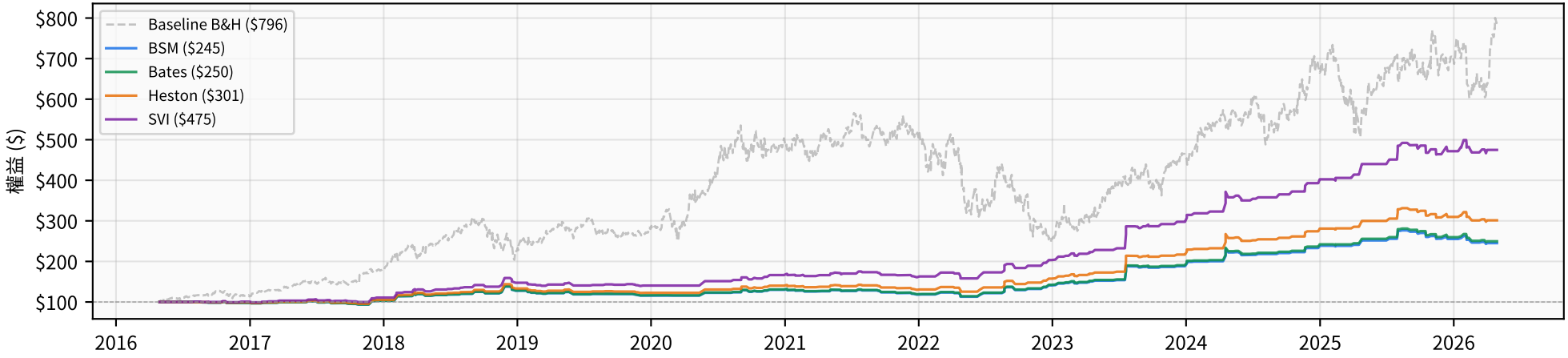
QQQ — 4 定價模型權益對比 ($100 起始)

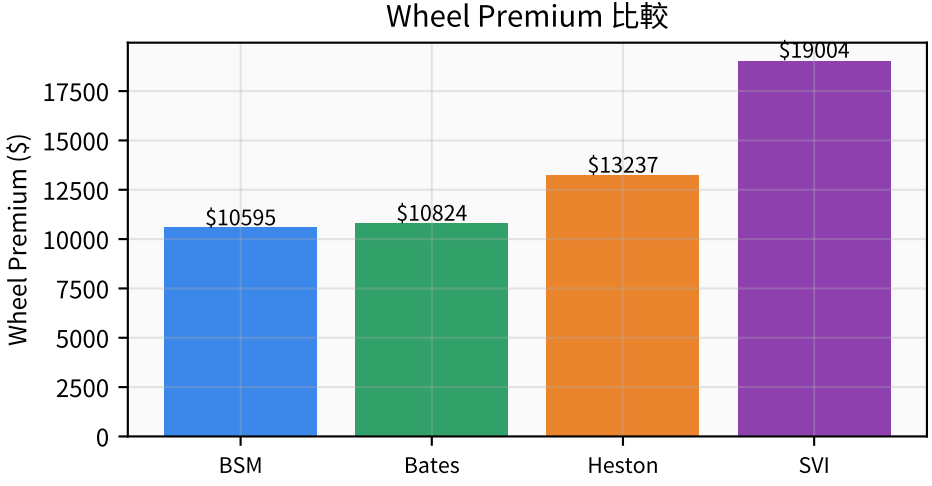


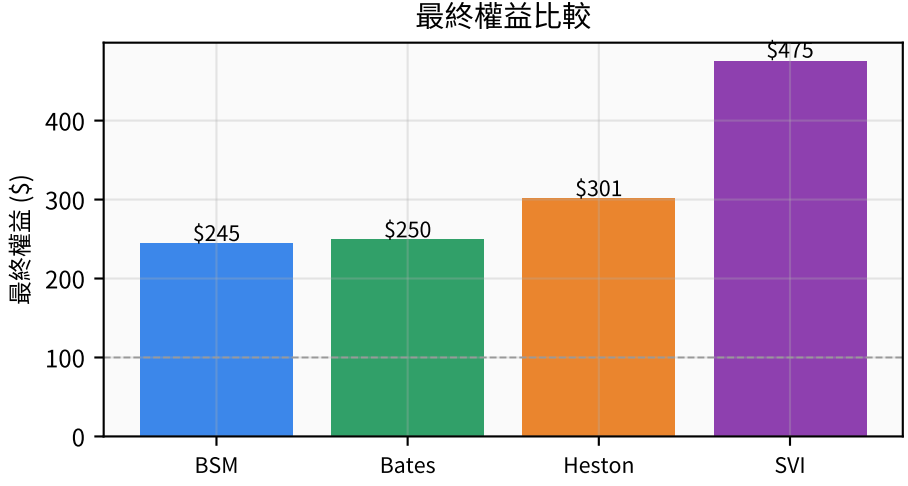




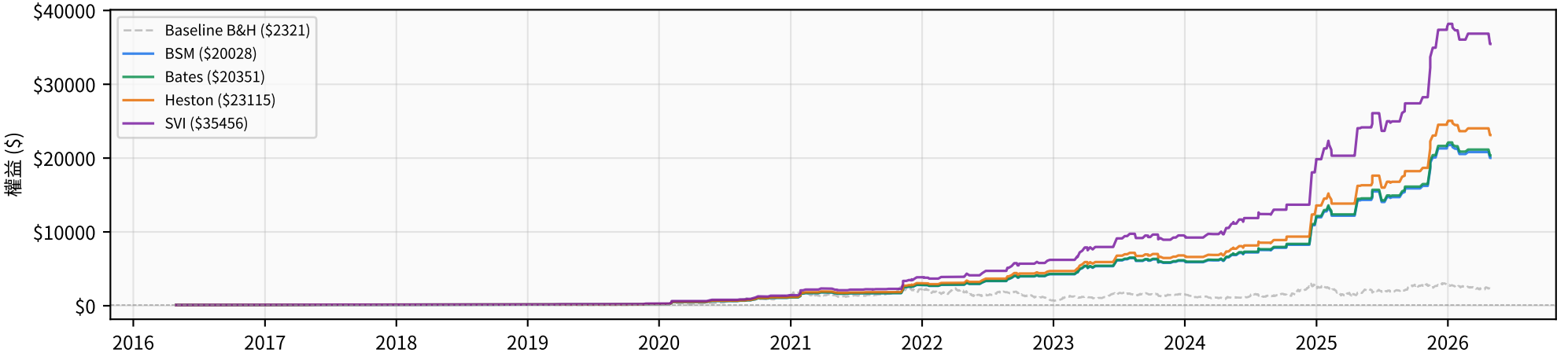
AMZN — 4 定價模型權益對比 ($100 起始)

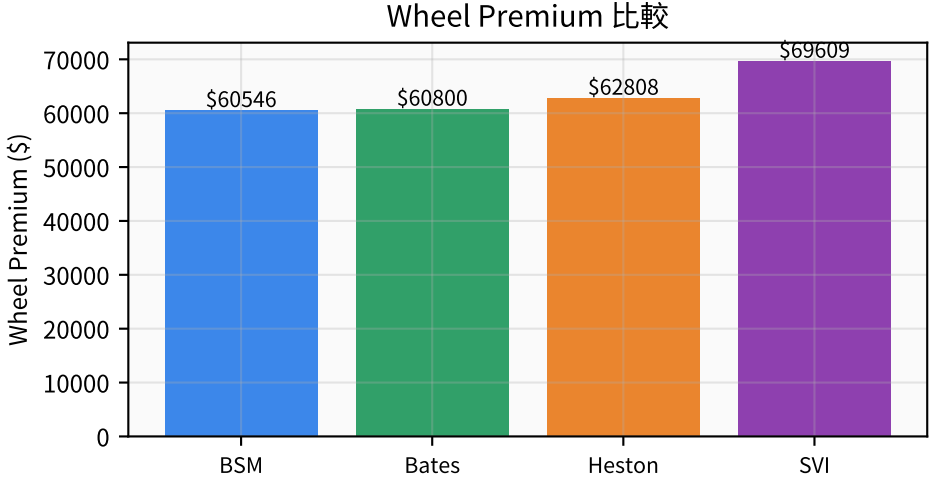


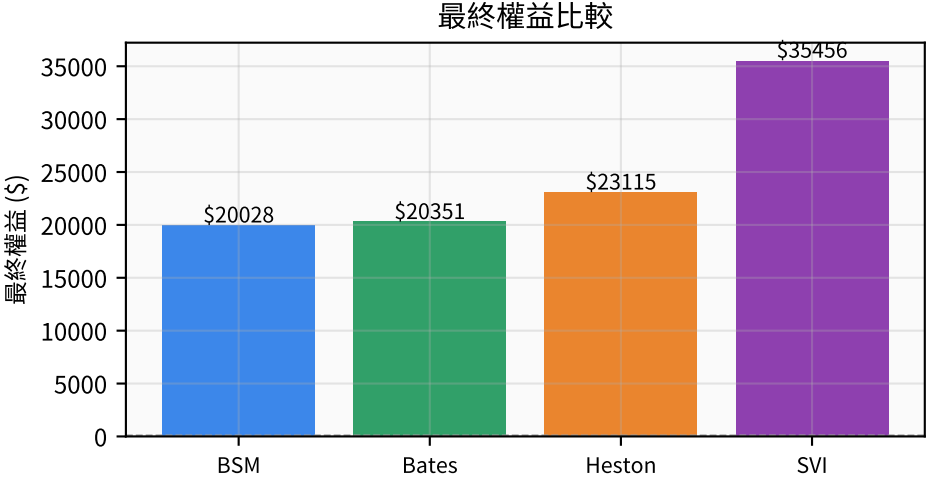




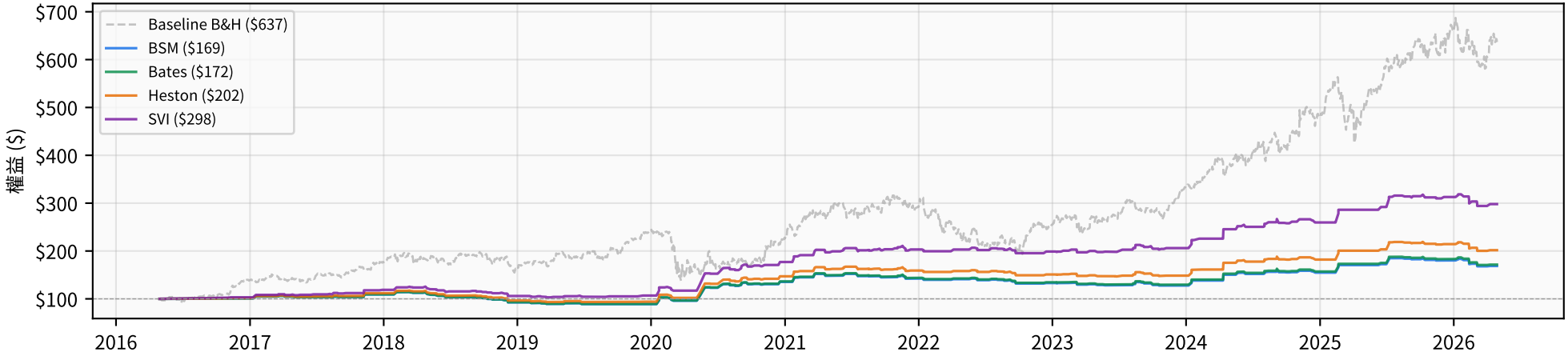
TSLA — 4 定價模型權益對比 ($100 起始)

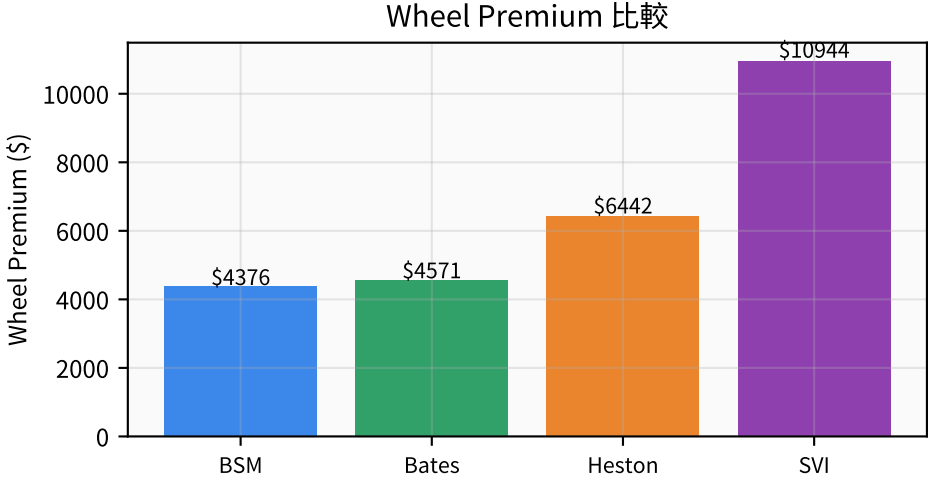


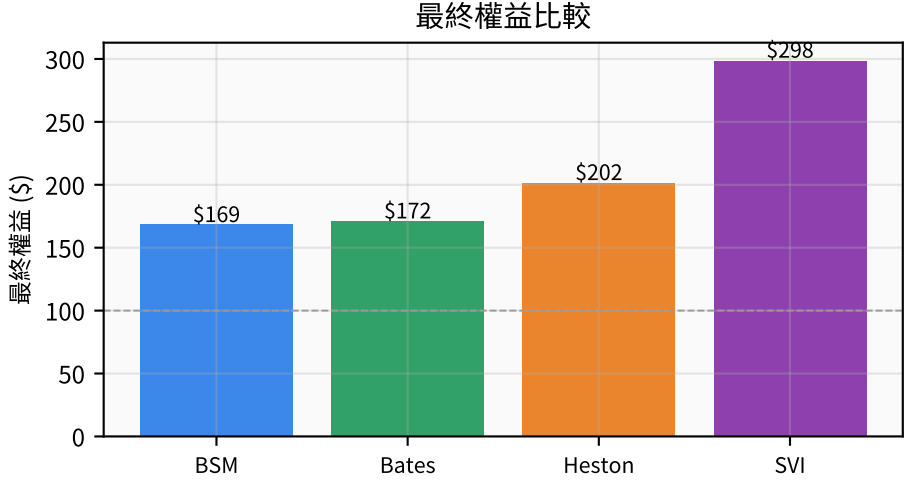




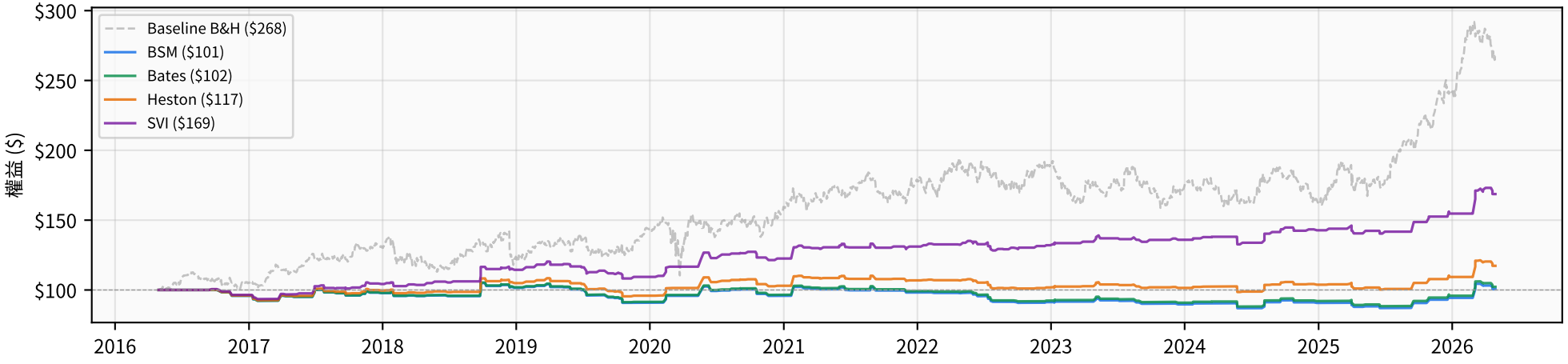
JPM — 4 定價模型權益對比 ($100 起始)

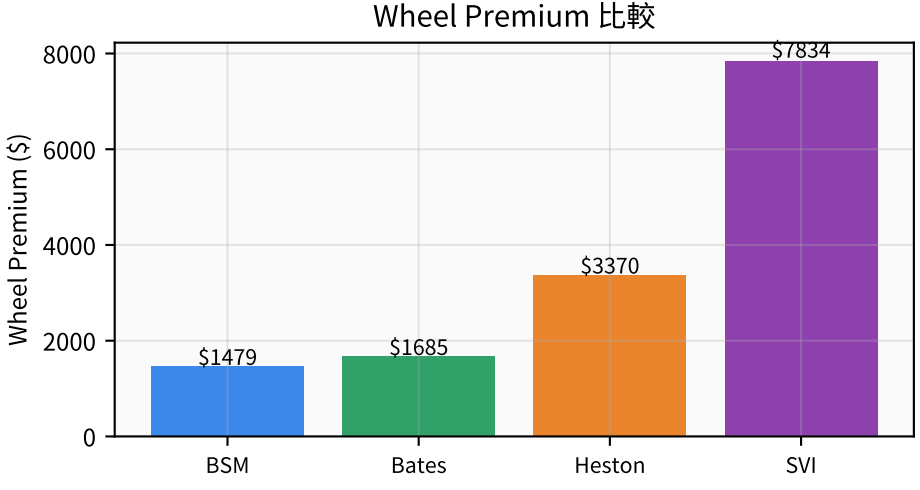


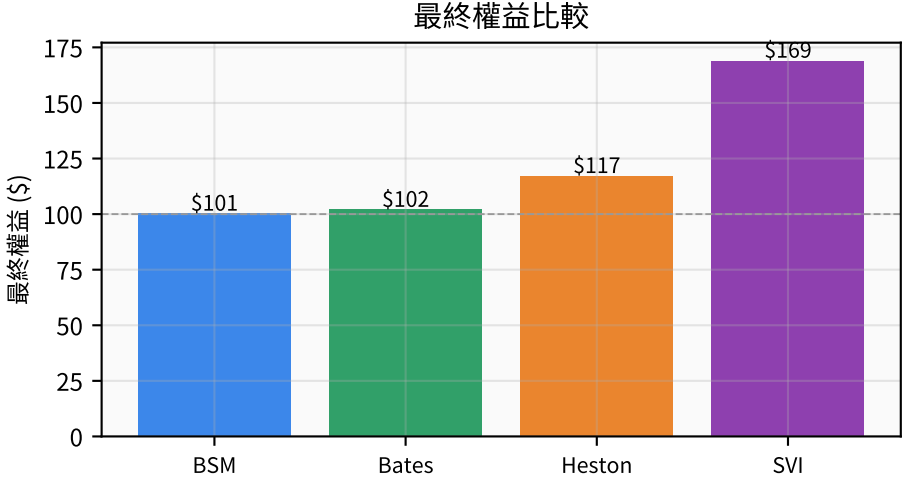




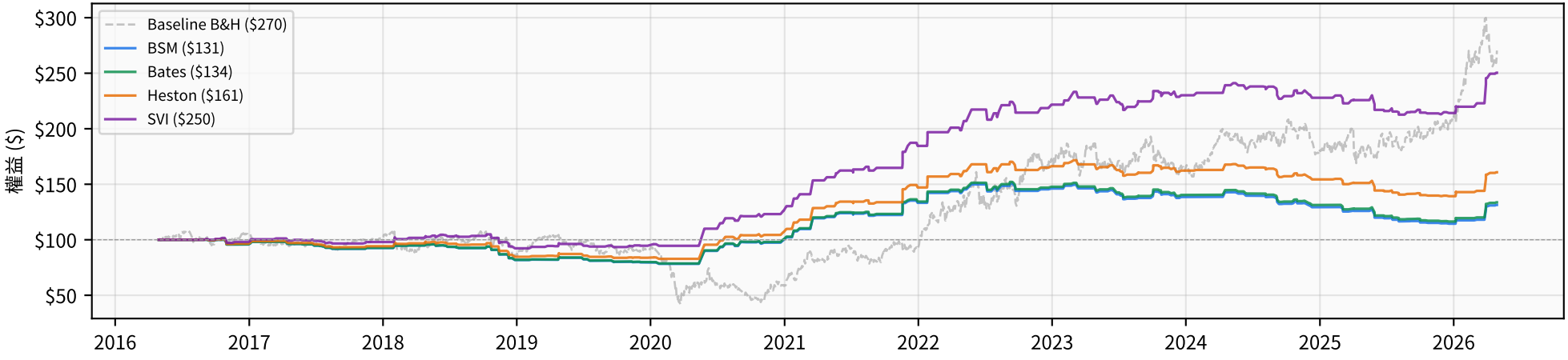
JNJ — 4 定價模型權益對比 ($100 起始)

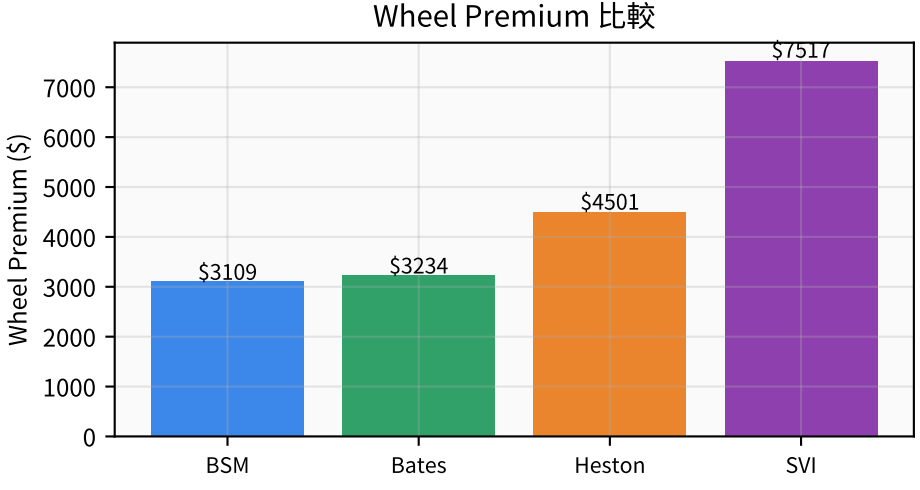


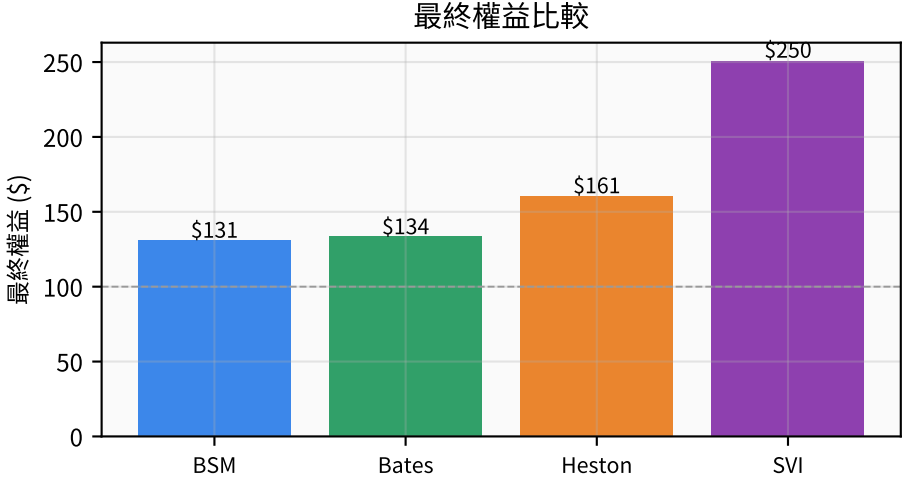




XOM — 4 定價模型權益對比 ($100 起始)







Wheel 詳情 — CSP/CC 週期及權利金

GOOGL

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	67	2	3	0	$ 5197
Bates	68	2	3	0	$ 5357
Heston	68	2	3	0	$ 7039
SVI	68	2	3	0	$ 10995

AAPL

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	61	1	1	0	$ 6004
Bates	61	1	1	0	$ 6166
Heston	61	1	1	0	$ 7861
SVI	61	1	1	0	$ 11835

MSFT

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	74	2	3	2	$ 11683
Bates	77	2	3	2	$ 12076
Heston	77	2	3	2	$ 15874
SVI	77	2	3	2	$ 25204

NVDA

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium

BSM	63	8	10	2	$ 8375
Bates	63	8	10	2	$ 8442
Heston	63	8	10	2	$ 8977
SVI	63	8	10	2	$ 10576

SPY

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	44	0	1	0	$ 4512
Bates	57	0	1	0	$ 4970
Heston	57	0	1	0	$ 8868
SVI	57	0	1	0	$ 18314

QQQ

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	66	0	2	0	$ 7706
Bates	73	0	2	0	$ 8201
Heston	73	0	2	0	$ 12689
SVI	73	0	2	0	$ 23795

AMZN

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	75	4	5	1	$ 10595
Bates	75	4	5	1	$ 10824
Heston	75	4	5	1	$ 13237
SVI	75	4	5	1	$ 19004

TSLA

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	74	18	19	6	$ 60546
Bates	74	18	19	6	$ 60800
Heston	74	18	19	6	$ 62808
SVI	74	18	19	6	$ 69609

JPM

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	68	0	1	0	$ 4376
Bates	69	0	1	0	$ 4571
Heston	69	0	1	0	$ 6442
SVI	69	0	1	0	$ 10944

JNJ

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	57	1	1	0	$ 1479
Bates	68	1	1	0	$ 1685
Heston	68	1	1	0	$ 3370
SVI	68	1	1	0	$ 7834

XOM

模型	CSP 週期	CC 週期	指派	被Call走	總 Premium
BSM	72	2	2	0	$ 3109
Bates	76	2	3	0	$ 3234

Heston	76	2	3	0	$ 4501
SVI	76	2	3	0	$ 7517

關鍵發現與建議

定價模型結論

最佳模型分佈：BSM=0 Bates=0 Heston=0 SVI=11

平均最終權益：
BSM 2295(*avgpremium*11235)
Bates 2332(*avgpremium*11484)
Heston 2662(*avgpremium*13788)
SVI 4044(*avgpremium*19602)

Bates 模型因跳躍項增加尾部風險定價，Wheel premium 通常 >= BSM。
Heston 透過偏態調整 OTM put 定價略高。
SVI 的波動率曲面可能對深 OTM 有不同表現。

策略結論

1/11 支標的 MultiMode 策略勝過 Buy-and-Hold

多模式引擎核心優勢：
1. 熊市自動切換 CASH 模式避免重大回撤
2. 盤整期透過 Wheel 產生額外選擇權收入
3. 牛市趨勢跟蹤 + 突破捕捉雙引擎並行
4. ATR-based 動態追蹤停損控制下行風險

部署建議

1. 定價模型選擇：
   - 一般用途推薦 BSM (最快、最穩定)
   - 高波動標的可考慮 Bates (更高 premium 但更準確)
   - 不建議單獨依賴 SVI (美元定價偏差較大)
2. 策略部署：
   - Mode B (+ Regime Filter) 為最低可行部署版本

- 趨勢型成長股效果最佳 (NVDA, GOOGL, AAPL)
- 防禦型/價值股效果較差 (JNJ, XOM)
- TSLA 等極高波動標的需額外風控

3. 風險管理：
- 建議最大單一標的曝險不超過 20%
- Wheel 部位需準備足夠保證金 (100 股 x 行權價)
- 定期檢視 regime 偵測器參數是否需調整

報告生成時間: 30 秒
報告日期: 2026-04-30 01:40